FUNCTORIALITY, COLIMIT AND REDUCTION PROPERTIES OF THE MAXIMAL WEAKLY ETALE SUBALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The counted statement span closes at the author's \end{lemma} on source line 30930. Its last enumerated item is the author's own placeholder '\item add more here.', which is delivered verbatim rather than removed and carries no mathematical condition. The ten references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The construction and universal property of the maximal weakly etale subalgebra B_max(A). invoked at statement lines 30914-30916 and used throughout the counted proof span. Is the defining property from which each listed assertion is deduced.
- Permanence of weakly etale maps under composition, base change and filtered colimits. used throughout the counted proof span. Supplies the stability that lets the maximal subalgebra be identified after each operation.

Source reference. Original lemma 00QO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 974–974. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9644–9644. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30504: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30504–30504. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 30846: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30846–30846. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30892: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30892–30892. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30454: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30454–30454. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let K be a field. For a K-algebra A denote $B_{max}(A)$ the maximal weakly étale K-subalgebra of A as in Lemma 30892. Then*

1. *any K-algebra map $A' \to A$ induces a K-algebra map $B_{max}(A') \to B_{max}(A)$,*
2. *if $A' \subset A$, then $B_{max}(A') = B_{max}(A) \cap A'$,*
3. *if $A = \operatorname{colim} A_i$ is a filtered colimit, then $B_{max}(A) = \operatorname{colim} B_{max}(A_i)$,*
4. *the map $B_{max}(A) \to B_{max}(A_{red})$ is an isomorphism,*
5. *$B_{max}(A_1 \times \ldots \times A_n) = B_{max}(A_1) \times \ldots \times B_{max}(A_n)$,*
6. *if A has no nontrivial idempotents, then $B_{max}(A)$ is a field and a separable algebraic extension of K,*
7. *add more here.*

Proof. Proof of (1). This is true because the image of $B_{max}(A') \to A$ is weakly étale over K by Lemma 30846.

Proof of (2). By (1) we have $B_{max}(A') \subset B_{max}(A)$. Conversely, $B_{max}(A) \cap A'$ is a weakly étale K-algebra by Lemma 30846 and hence contained in $B_{max}(A')$.

Proof of (3). By (1) there is a map $\operatorname{colim} B_{max}(A_i) \to A$ which is injective because the system is filtered and $B_{max}(A_i) \subset A_i$. The colimit $\operatorname{colim} B_{max}(A_i)$ is weakly étale over K by Lemma 30454. Hence we get an injective map $\operatorname{colim} B_{max}(A_i) \to B_{max}(A)$. Suppose that $a \in B_{max}(A)$. Then a generates a finitely presented K-subalgebra $B \subset B_{max}(A)$. By Algebra, Lemma 00QO there is an i and a K-algebra map $f : B \to A_i$ lifting the given map $B \to A$. Since B is weakly étale by Lemma 30846, we see that $f(B) \subset B_{max}(A_i)$ and we conclude that a is in the image of $\operatorname{colim} B_{max}(A_i) \to B_{max}(A)$.

Proof of (4). Write $B_{max}(A_{red}) = \operatorname{colim} B_i$ as a filtered colimit of étale K-algebras (Lemma 30504). By Algebra, Lemma 07K4 for each i there is a K-algebra map $f_i : B_i \to A$ lifting the given map $B_i \to A_{red}$. It follows that the canonical map $B_{max}(A_{red}) \to B_{max}(A)$ is surjective. The kernel consists of nilpotent elements and hence is zero as $B_{max}(A_{red})$ is reduced (Lemma 30846).

Proof of (5). Omitted.

Proof of (6). Follows from Lemma 30846 part (4). □